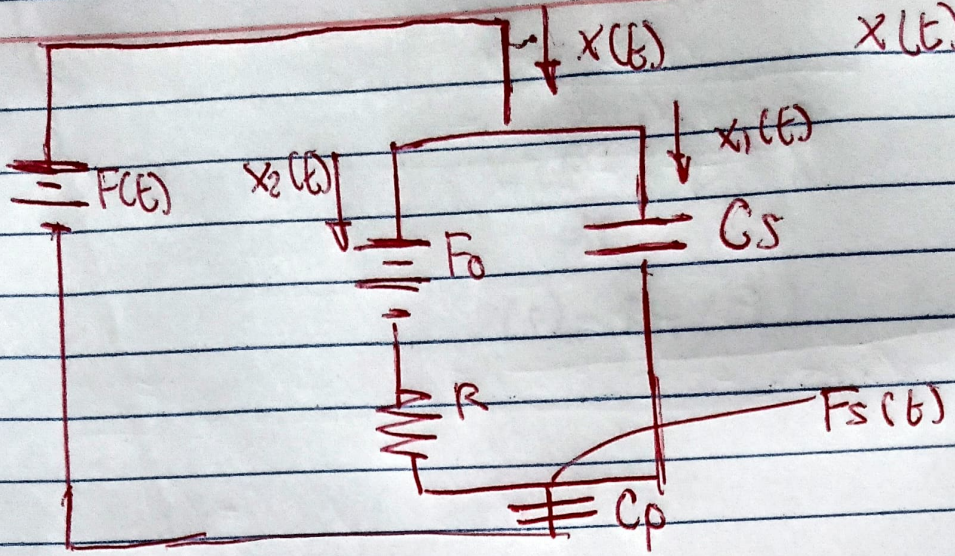
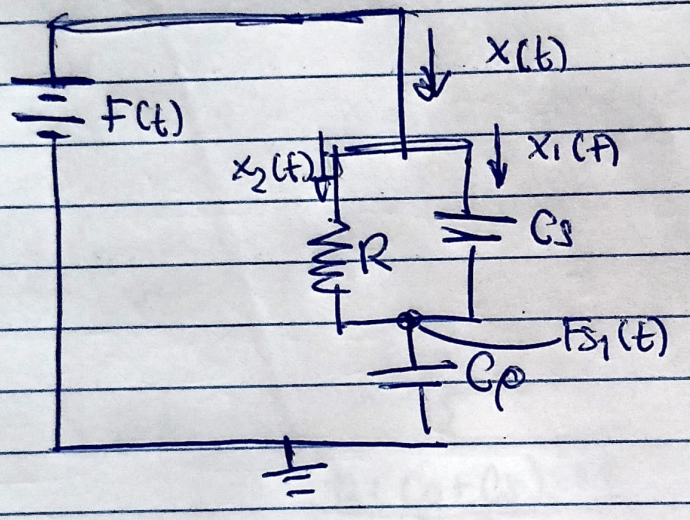
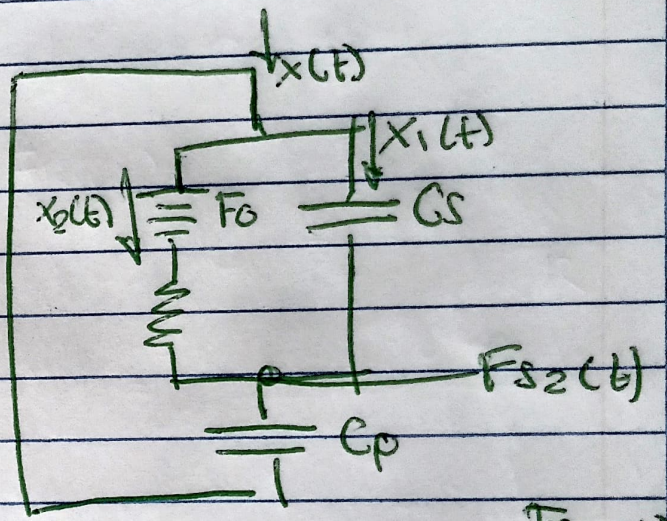
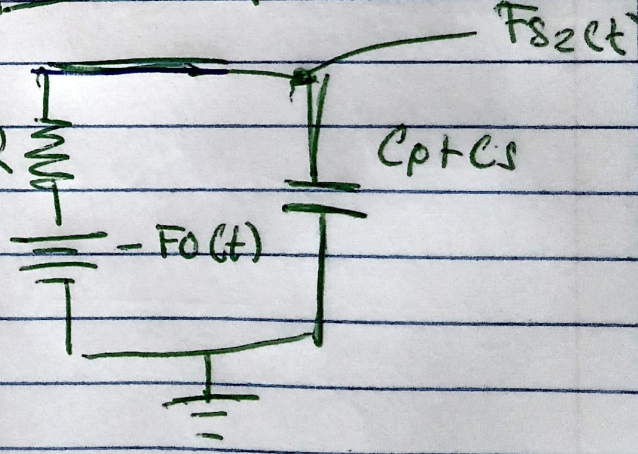
$x(t) = x_1(t) + x_2(t)$









Análisis Matemático

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

Circuito 1: Apagando Fo(t)

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d(F(t) - F_{s1}(t))}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{\alpha R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d(F(t) - F_{s1}(t))}{dt} + \frac{F(t) - F_{s1}(t)}{\alpha R}$$

$$C_p s F_{s1}(s) = C_s s (F(s) - F_{s1}(s)) + \frac{F(s) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$F_{s1}(s)\left(C_p s + C_s s + \frac{1}{R}\right) = F(s)\left(\frac{1}{R} + C_s s\right)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}}$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s) + 1} F(s)$$

Circuito 2: Apagando $F(t)$

Recordando que $F_o(t) = \alpha F(t)$

$$-\alpha F(t) = R(x(t)) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s} \qquad F(s) = \frac{-1}{\alpha} R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$\frac{F_{s+}(s)}{F(s)} = \frac{\frac{x(s)}{C(C_p + C_s)s}}{-\frac{1}{\alpha}\left(\frac{R x (C_p + C_s)s + 1}{(C_p + C_s)s} x(s)\right)}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = \frac{\alpha F(s)}{RC(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s)$$

$$F_s(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{RC(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{RC(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{RC(C_p + C_s)s + 1}$$

$$\frac{Fs(s)}{Fs} = \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} \; sF(s)\left(1 - \frac{Fs(s)}{Fs}\right)$$

$$e(s) = \lim_{s \to 0} \; s\frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right)$$

$$= 1 - 1 + \alpha = \alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = \emptyset$$

$$\lambda = -1/RC_p + RC_s$$

El sistema es estable